

Power BI Concept Application

1. Import Mode vs DirectQuery Mode

Import mode copies data into the PBIX file, making reports fast because visuals use in-memory data. Refresh is required to see source changes.

DirectQuery does not store data; every visual sends a query to the source database, showing live data but depending on database performance.

Example:

- Static CSV → Use Import Mode.
- SQL Server with live transactions → Use DirectQuery.

Explore:

Home → Get Data → Text/CSV (Import).

Note that the visuals work even if disconnected after loading because data is stored in PBIX.

2. Text Data Type on Card Visual

A Card visual can only aggregate numeric fields. If Sales is stored as Text, Power BI cannot calculate SUM and may show Count or Don't Summarize.

Fix in Power Query (recommended):

Home → Transform Data → Select column → Transform → Data Type → Whole Number/Decimal Number.

Alternative:

Or you can change the datatype of your column by: Data View → Column tools → Data Type.

Example:

'100','200','300' stored as text cannot be summed until converted to numeric.

3. Change Bar Chart Sort Order

Select the bar chart.

Click the three dots (...) at the top-right of the visual.

Choose Sort by → Sales (or required field) → Descending or Ascending

Sort By will show both Categories and Numerical Columns.

Example:

Alphabetical: Ahmedabad (300), Delhi (900), Mumbai (700).

Descending by Sales: Delhi (900), Mumbai (700), Ahmedabad (300).

4. Power Query Filter vs Top N Filter

Power Query filtering removes rows (permanently from the dataset) before data enters the model. Every report visual uses the filtered dataset.

Top N filter only affects the selected visual while other visuals continue using the complete dataset. It doesn't remove rows, just filter them temporarily for specific visual or Page.

Explore:

Power Query: Home → Transform Data.

Top N: Select visual → Filters Pane → Visual Level Filters → Top N → Enter value → Apply Filter.

5. Show Percentage of Total Sales

For example, we imported a csv file with columns such as Sales and Region.

Create measures:

Total Sales = SUM(Sales[Sales])

Sales % = DIVIDE([Total Sales], CALCULATE([Total Sales], ALL(Sales[Region])))

Format the measure as Percentage using Measure Tools. When you click on a measure, a new ribbon named Measure Tools will pop-up. There we can change the format of our measure.

Use Region on Axis and Sales % in Values.

Example:

North=400, South=300, East=200, West=100

Chart shows 40%, 30%, 20%, 10% instead of absolute values.

6. Hardcoded File Paths

Power BI stores the exact file path of imported files.

Example:

C:\Users\Shivam\Desktop\Sales.csv

If another user opens the PBIX and the file is missing from that location, refresh fails.

View the source path:

Home → Transform Data → Advanced Editor.

Make reports portable by:

- Using SharePoint/OneDrive/SQL Server.
- Using Power Query Parameters. (In Transform Data)
- Keeping a common folder structure for all users.